UNIT 3 · TOPIC 1(C)

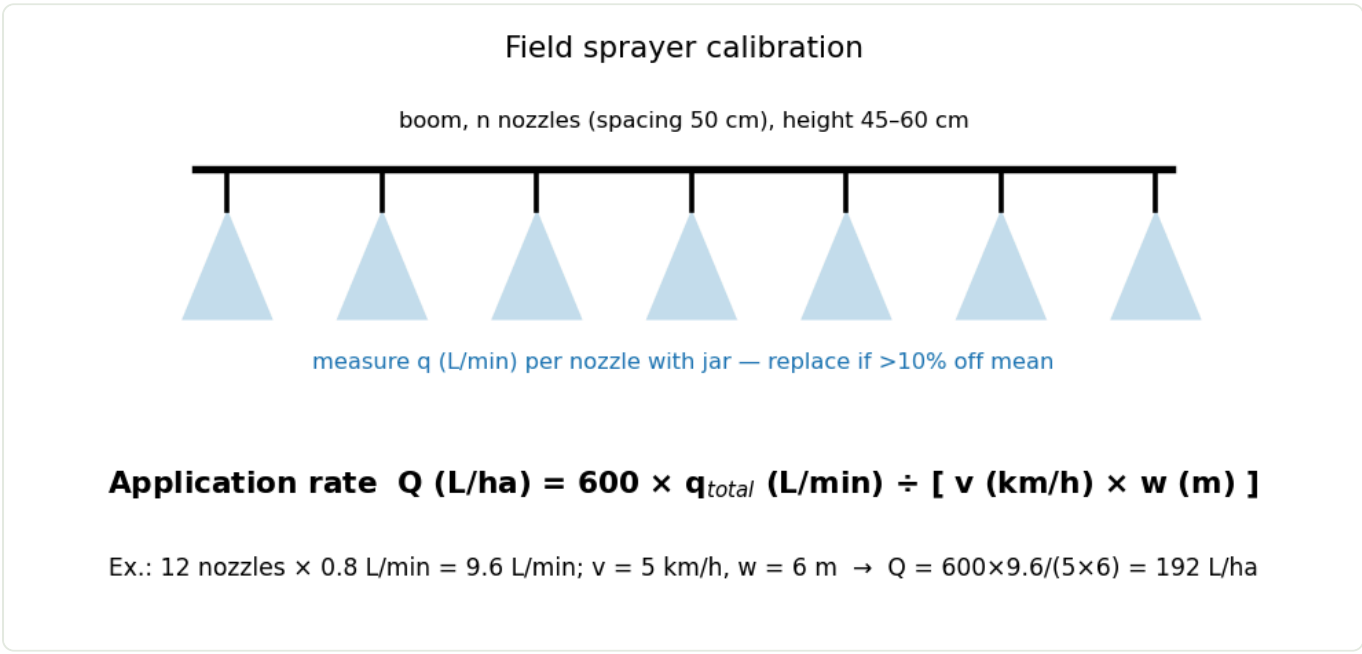
Calibration of Plant-Protection Equipment

Syllabus topic: Calibration of seed drills, planters and plant protection equipment

Field sprayer calibration

1. Fill the tank with clean water and operate at working pressure.
2. Collect the discharge of EACH nozzle in a jar for 1 min — q (L/min). Replace any nozzle deviating **>±5%** from the mean, or **>10% above rated** (worn tip).
3. Measure the actual forward speed over a marked **30–50 m** run.
4. Compute the application rate (formula below).
5. Adjust speed, nozzle size or (slightly) pressure to reach the recommended rate.
6. Verify: check the tank volume used over a measured area.

$$Q\ (\text{L/ha}) = 600 \times \text{total discharge (L/min)} \div [\text{speed (km/h)} \times \text{effective swath (m)}]$$



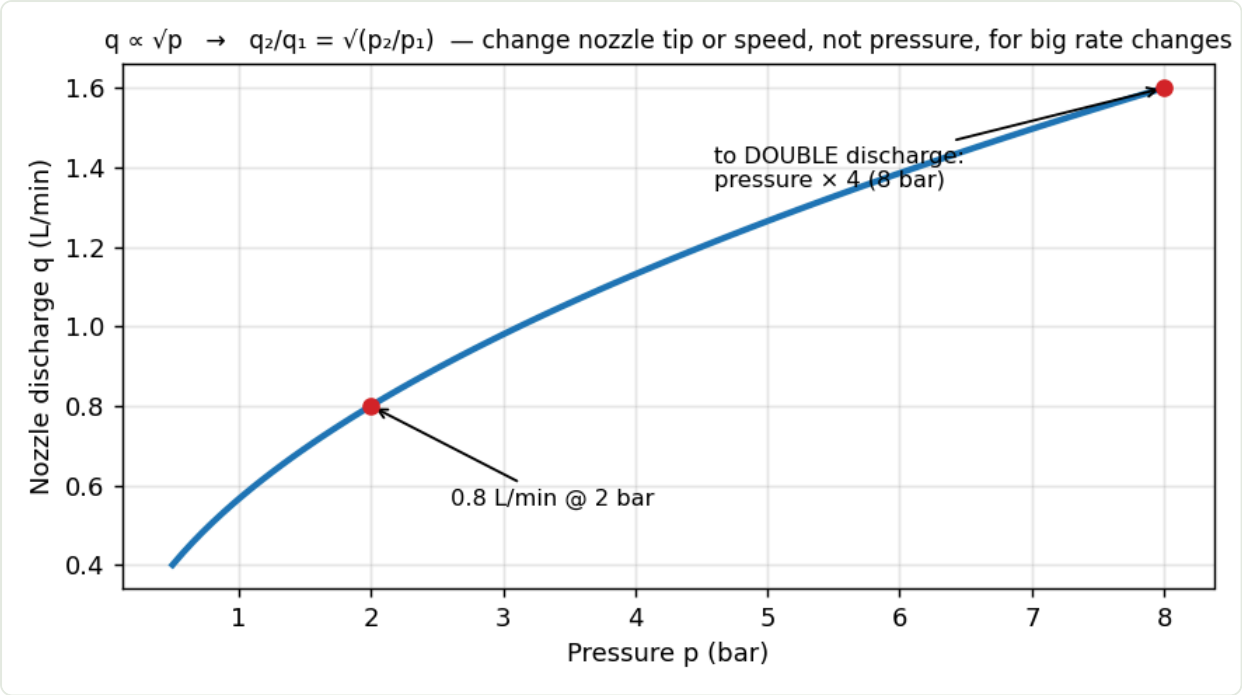
Field sprayer calibration (the figure abbreviates the replacement rule — use step 2 above)

Speed check: 30 m covered in 21.6 s → v = 30/21.6 = 1.39 m/s = 5 km/h.

Pressure–discharge relation

$$q \propto \sqrt{p} \rightarrow q_2/q_1 = \sqrt{(p_2/p_1)}$$

- Nozzle flow is orifice flow. Doubling the discharge needs **FOUR times** the pressure — which also shrinks droplets and raises drift.
- So make large rate changes by changing nozzle tips or forward speed; use pressure only for fine trimming.
- Pressure also sets droplet size: higher p → finer drops.



Nozzle discharge vs pressure

Knapsack sprayers and dusters

- **Knapsack (refill method):** spray a measured area (e.g., 10 m × swath) at normal walking speed (**~3 km/h**) and constant pressure; measure the volume used → L/ha. The operator's pace and nozzle-height discipline dominate accuracy.
- **Dusters/granular applicators:** run for a set time or distance at a set metering opening; catch and WEIGH the output → kg/min. Combine with speed and swath — the same logic as sprayers, by mass:

$$\text{Rate (kg/ha)} = 600 \times \text{(kg/min)} \div (\text{v} \times \text{w})$$

Typical values and nozzle selection

- Boom nozzle spacing **50 cm** at **45–60 cm** height (flat-fan, 30–50% overlap).
- **Herbicides:** FLAT-FAN nozzles at low pressure (**1–3 bar**) for uniform band/blanket coverage.
- **Insecticides/fungicides:** HOLLOW-CONE nozzles at higher pressure (**3–7 bar**) for canopy penetration.

Spraying	Application rate
High volume	>400 L/ha
Conventional field spraying	Commonly 200–500 L/ha (knapsack ~400–500 L/ha)
Low volume	50–200 L/ha
ULV (special rotary atomisers)	<5 L/ha
Drone ULV	20–40 L/ha (Unit 2, Topic 12)

Practice MCQs

Q19. Sprayer application rate formula is Q (L/ha) =:

(a) 600q/(v·w) (b) q·v·w/600
(c) 60q/(v·w) (d) 1000q/(v·w)

Q22. Nozzle discharge varies with pressure as:

(a) $q \propto p$ (b) $q \propto \sqrt{p}$
(c) $q \propto p^2$ (d) Independent

Q23. To double nozzle discharge, pressure must be increased:

(a) 2× (b) 3×
(c) 4× (d) 8×

Q24. Large changes in application rate are best made by:

(a) Pressure alone (b) Changing nozzle tips or forward speed
(c) Adding water (d) Raising the boom

Q25. A nozzle should be replaced when its discharge exceeds the rated value by more than:

(a) 2% (b) 5%
(c) 10% (d) 25%

Q26. Increasing spray pressure makes droplets:

(a) Coarser (b) Finer (increasing drift risk)
(c) Unchanged (d) Heavier

Q27. For herbicide application the preferred nozzle is:

(a) Hollow cone (b) Flat fan at low pressure
(c) Solid stream (d) Rain gun

Q28. For insecticide/fungicide canopy penetration the common nozzle is:

(a) Flat fan (b) Hollow cone
(c) Flooding (d) Anvil

Q29. Typical boom nozzle spacing and height are:

(a) 25 cm, 20 cm (b) 50 cm, 45–60 cm
(c) 100 cm, 1.5 m (d) 75 cm, 10 cm

Q30. Knapsack sprayer calibration by the refill method measures:

(a) Pressure (b) Volume used over a known sprayed area
(c) Droplet size (d) Pump wear

Q31. Duster output for calibration is determined by:

(a) Volume in a jar (b) Weighing output per unit time
(c) Colour (d) Sound

Q32. Ultra-low-volume (ULV) spraying refers to application rates:

(a) <5 L/ha (b) 50–100 L/ha
(c) 200 L/ha (d) >400 L/ha

Answers

Q19 — (a) 600q/(v·w)
Q22 — (b) $q \propto \sqrt{p}$
Q23 — (c) 4× — *$q \propto \sqrt{p} \rightarrow$ for $q_2/q_1 = 2$, $p_2/p_1 = 2^2 = 4$.*
Q24 — (b) Changing nozzle tips or forward speed
Q25 — (c) 10%

Q26 — (b) Finer (increasing drift risk)
Q27 — (b) Flat fan at low pressure
Q28 — (b) Hollow cone
Q29 — (b) 50 cm, 45–60 cm
Q30 — (b) Volume used over a known sprayed area
Q31 — (b) Weighing output per unit time
Q32 — (a) <5 L/ha

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting) & Ch. 21 (Spraying and Dusting); Jagdishwar Sahay — Elements of Agricultural Engineering; ISI/BIS & RNAM test codes for seed drills and sprayers; NPTEL Farm Machinery lectures.